

Week 4 Task Report

This week's task was all about **getting started with Python for data visualization**. Compared to Week 1, which introduced the basics of data and visualization concepts, Week 2 was more technical and hands-on. I had to set up my environment, work with a dataset, and then generate different types of plots to see how raw data can be transformed into meaningful visuals. It felt like the first real step into practical visualization, where I wasn't just learning theory but actually producing charts and graphs that tell a story.

Part 1: Environment Setup

I worked in **Jupyter Notebook** and prepared the environment for visualization. This included installing and importing the necessary libraries. The main tools I used were:

- **Pandas** for handling and manipulating data.
- **Matplotlib** for creating charts and graphs.
- **Seaborn** for more advanced and visually appealing plots.

This setup gave me the foundation to start experimenting with different types of visualizations.

Part 2: Data Import

The dataset I worked with contained **YouTube channel statistics** related to the World Cup. It included details such as:

Video Name

View Count

Comment Count

Channel Name

Published Date

Like Count

Video Duration

Exploring the dataset helped me understand its structure and decide which variables to use for different plots.

Part 3: Basic Plots with Matplotlib

I created several types of plots to practice visualization:

1. Line Chart – Trend Over Time

- Showed how view counts changed over time for a selected channel.
- Helped identify peaks and dips in audience engagement.

2. Bar Chart – Category Comparison

- Compared view counts across different channels.
- Made it easy to see which channels had the highest engagement.

3. Pie Chart – Proportions

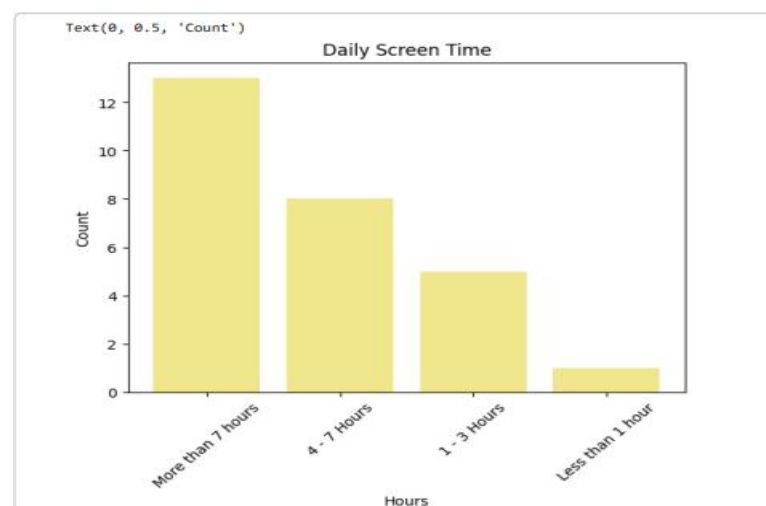
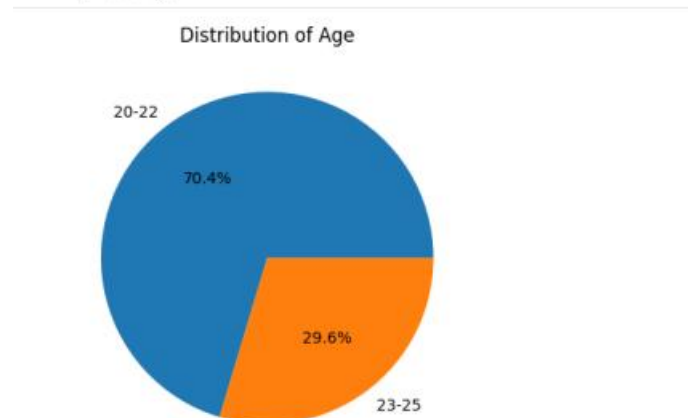
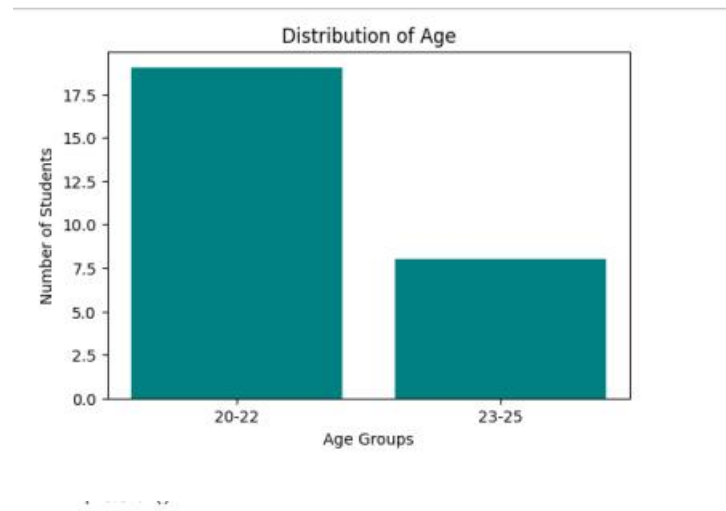
- Displayed the percentage share of views by channel.
- Highlighted which channels dominated overall viewership.

4. Scatter Plot – Relationships

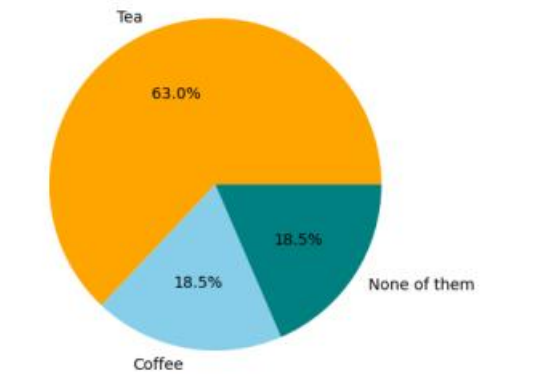
- Plotted likes against views to check for correlation.

- Showed how audience interaction (likes) related to overall popularity (views).

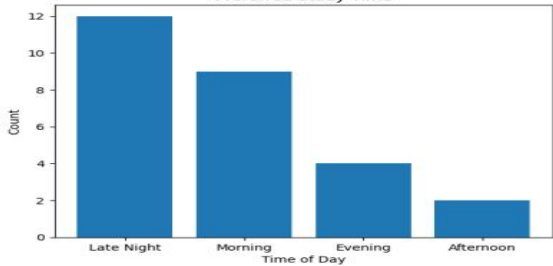
Output Screenshots:



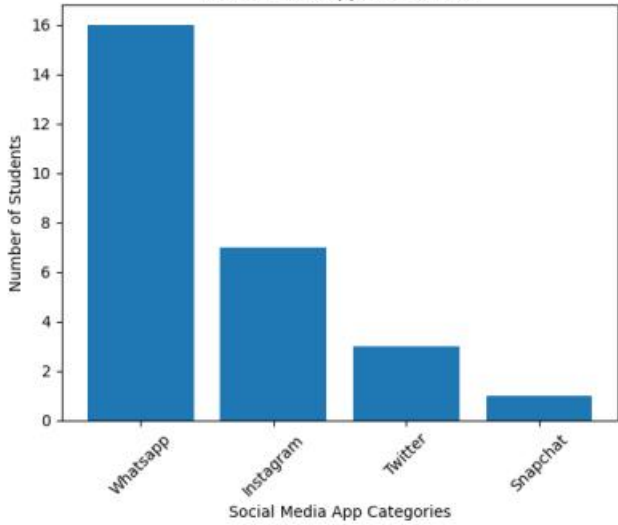
Coffee or Tea Preference



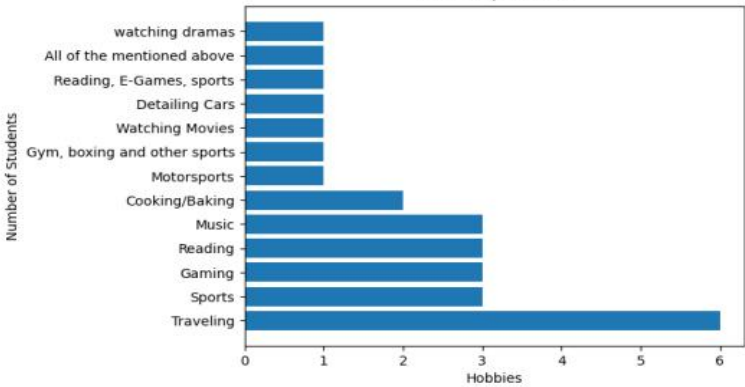
Preferred Study Time

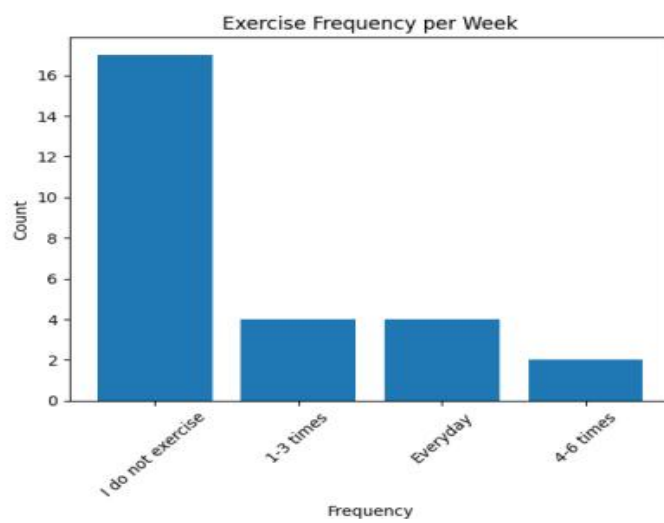
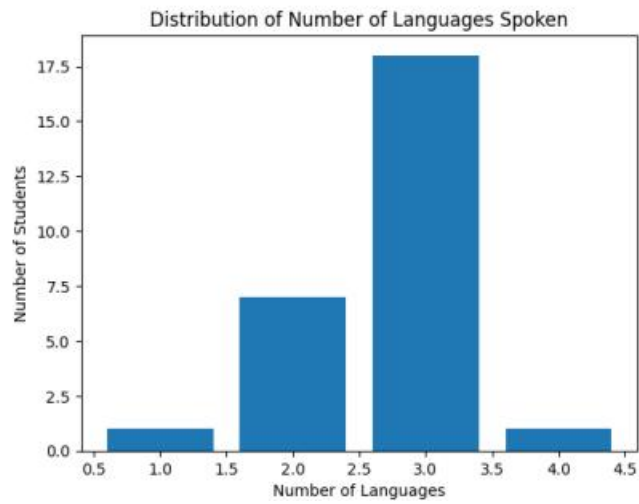


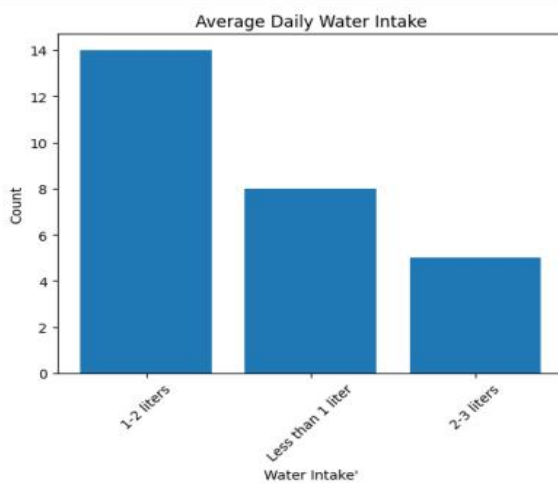
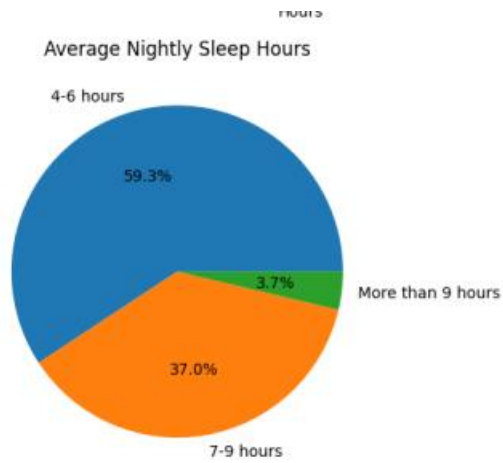
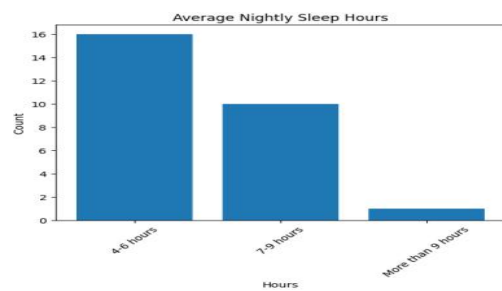
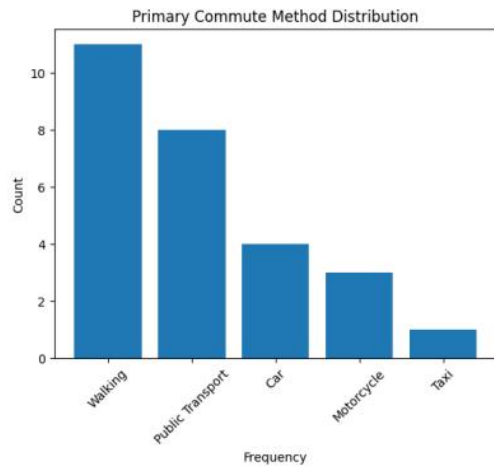
Social Media App Preferences

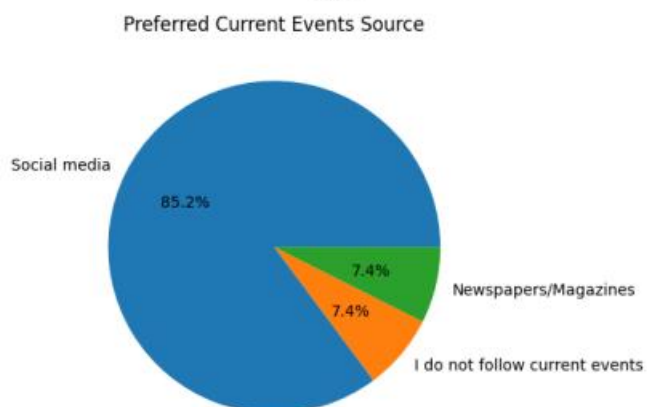
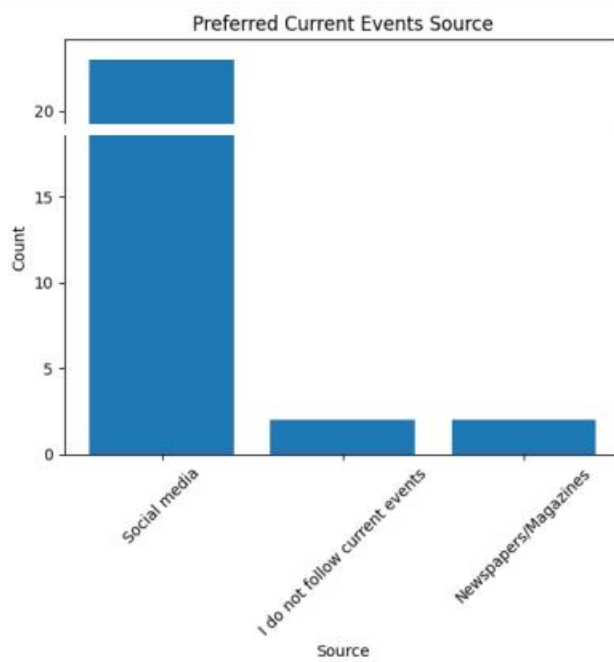
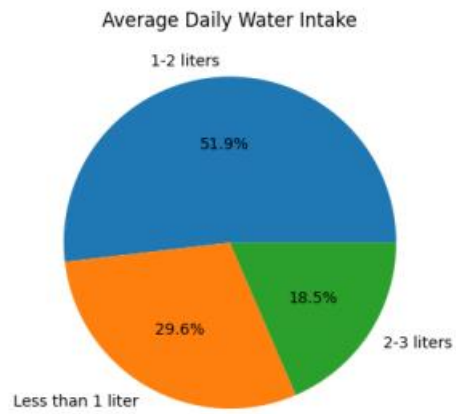


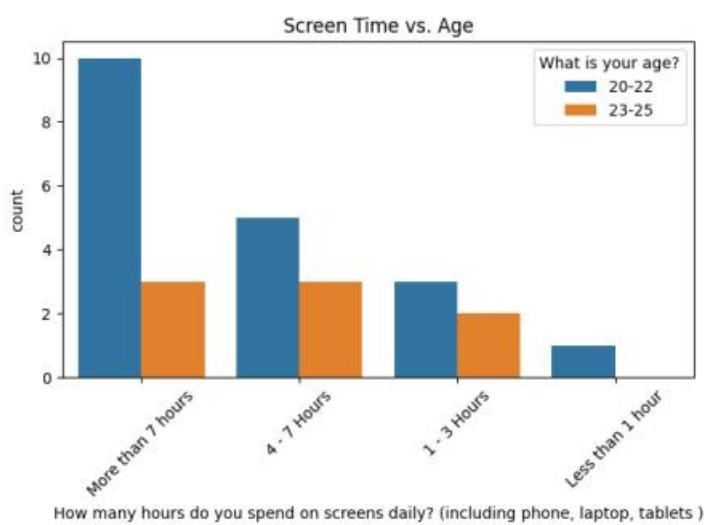
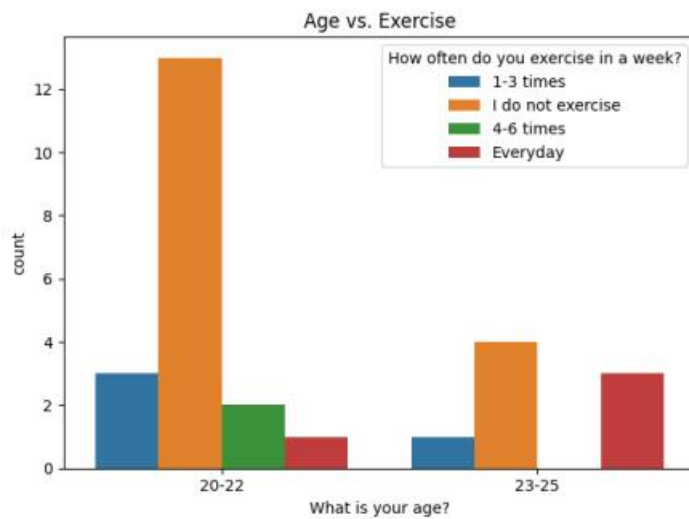
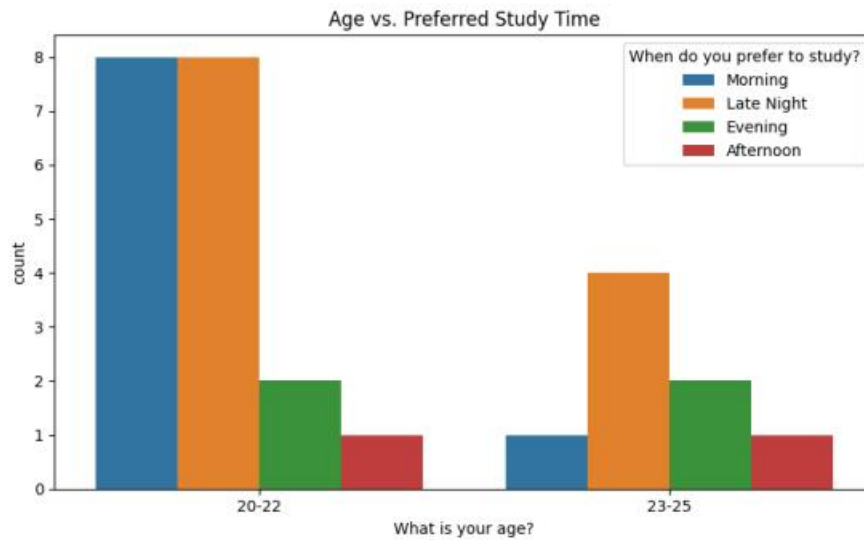
Hobbies / interest

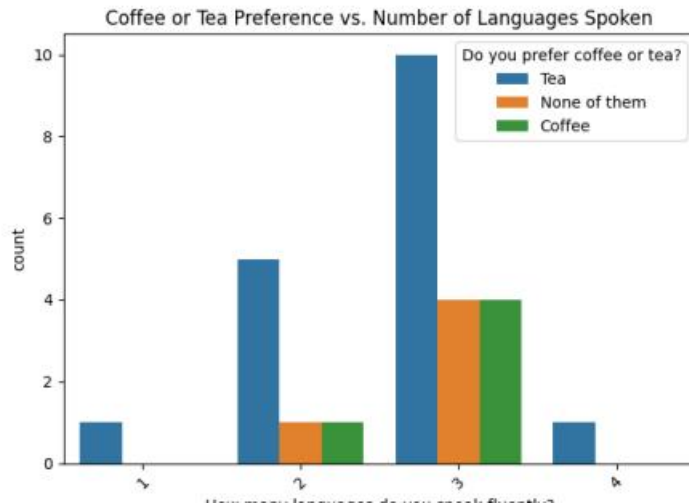












Part 4: Customization

I customized the charts to make them more professional and easier to interpret:

- Added **titles** and **axis labels** for clarity.
- Adjusted **colors** to differentiate categories.
- Rotated **x-axis ticks** to improve readability.
- Experimented with different chart styles to see how presentation affects understanding.

Part 5: Insights Summary

From the visualizations, I learned:

- **Line charts** are best for showing **trends over time**.
- **Bar charts** clearly compare **categories**.
- **Pie charts** highlight **proportions** within a dataset.
- **Scatter plots** reveal **relationships** between variables.

Customization is important, it makes visuals not only attractive but also easier to interpret.